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Explaining employment and environmental degradation nexus with environmental employment curve (EEC): A sector-wide threshold estimation for China

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ABSTRACT

China is the world's largest emitter of carbon dioxide and has experienced significant economic growth in the last decade. The country also has significant employment capacity and has witnessed rapid economic growth in the recent years. This study aims to investigate the existence of the environmental employment curve (EEC) in China's main economic sectors. This study examines the relationship between carbon emissions, economic growth, energy consumption, renewable energy consumption and employment (industry, services, agriculture) in China over the period 1990–2020 using threshold autoregressive regression estimation. According to the EEC hypothesis, there exists an inverted U-shaped relationship between environmental pollution and employment rate. Empirical results show that the EEC hypothesis seems to hold in China. Further, an analysis of the impact of employment on environmental pollution in China's three main sectors – industry, service and agriculture – shows that the EEC hypothesis cannot be rejected in the industrial and the agricultural sectors. In addition, empirical results show that a 1% increase in energy consumption leads to a 0.92% increase in carbon emissions, while the use of renewable energy leads to a 0.18% decrease. These results suggest that investments in green energy transformation do not cause employment loss. Hence, it is important to drive policies that promote investments in renewable and clean technologies, stimulate research in renewable energy-related institutions and support public-private partnerships. Additionally, to generate employment without compromising on the environment or prosperity, policymakers should choose to implement alternative technologies that are less polluting or do not produce any carbon emissions.

1. Introduction

Environmental pollution is a global concern and one of the leading causes of climate change. The lack of clear regulations related to environmental impacts is one of the key contributors to this problem. If the level of environmental pollution increases in one country, its negative effects are felt not only in that country but also in other parts of the world. This makes environmental pollution a widespread global crisis. The issue is compounded by the need to rein in environmental pollution without adversely affecting income. As the world population increases, it becomes imperative to provide employment opportunities to the newly added workforce. New employment opportunities are created through economic growth/increase in income. Simultaneously, an increase in employment levels boosts the total income of the people. In other words, employment contributes to economic growth/income and vice versa. However, studies shows that increase in economic growth is accompanied by a rise in pollution levels. Given this scenario, the main goal of this study is to explore the relationship between employment and environmental pollution.

Conventional environmental literature has shown the existence of a nexus between environmental pollution and economic growth/income. Moreover, many environmental studies have empirically confirmed the presence of a positive relationship between environmental pollution and income/economic growth (Alola et al., 2023; Kostakis and

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Arauzo-Corad, 2023; Nasrullah et al., 2023; Rahman and Kashem, 2017). The inverted "U" shaped relationship between environmental literature as the Environmental Kuznets Curve (EKC). According to the EKC hypothesis, there is a positive relationship between economic growth and environmental pollution until at least one turning point. Thereafter this turning point, as economic growth continues to increase, environmental pollution begins to decrease depending on the increase in welfare level. In many studies in the traditional environmental literature, the validity of the EKC hypothesis has been examined for several cases. However, recent scientific evidence from several countries have demonstrated attempts to explain the possibility of achieving a decoupling between economic performance improvement and environmental degradation (Andreoni and Galmarini, 2012; Apeaning, 2021; Chen et al., 2018; Dong et al., 2021; Engo, 2019; Gao et al., 2021; Hossain and Chen, 2020; Huang et al., 2021; Karakaya et al., 2019; Karmellos et al., 2021; Leitão et al., 2022; Naqvi, 2021; Papież et al., 2021; Rao et al., 2022; Raza and Lin, 2020; Shi, 2020). This implies that while economic growth continues to increase, environmental pollution, in the form of emissions, decreases.

Illustratively, the labour market vis-à-vis employment which remains a critical aspect an economy could then be associated with environmental sustainability aspect. According to Okun's law (1962), there exists a negative relationship between unemployment and the production/income of a country. That is, the greater the unemployment in a country, the greater the gap between the country's potential gross domestic product (GDP) and actual GDP. The GDP of an economy is, therefore, positively linked to both employment and environmental pollution. Kashem and Rahman (2020) empirically investigated the relationship between unemployment and environmental degradation and reported an inverse relationship between the two. Similar results have been reported by researchers in different countries (Anser et al., 2021; Bhowmik et al., 2021; Ng et al., 2022; Rayhan et al., 2020; Shang and Xu, 2022; Shastri et al., 2022; Tanveer et al., 2022).

Based on the literature cited above, considering all these relationships, it indicates the possibility that environmental pollution and employment are positively related. It is reasonable to assume that there is a positive relationship between employment and environmental pollution, especially when we consider the positive relationship between GDP and environmental pollution and the positive relationship between employment and GDP. We refer to this relationship as the environmental employment curve (EEC). To empirically verify this hypothesis, we analyse data from China, which accounts for the largest share of both carbon emissions and economic activity in the world. The main contribution of this research to the literature is revealing whether our hypothesis is empirically proven with the available data. Mainly this study contributes to the extant literature in the following ways: (i) it presents an entirely novel idea developed by the authors within the field of environmental economics; (ii) it is the first of its kind to explore the relationship between employment and environmental pollution; (iii) the empirical analyses confirm the EEC hypothesis in China; (iv) it facilitates a better understanding of the environmental-economic linkage by examining the impact of sector-specific employment (industrial, service and agricultural) on environmental pollution; and (v) finally, this study presents a set of policy recommendations on the basis of its empirical findings.

The rest of this paper is organised as follows. The next section (2) describes the dataset and the empirical approaches. Section 3 discusses the research findings in detail and links them to the literature. Finally, Section 4 presents the study's conclusions and policy effects.

2. Data and methodology

The description of the dataset and the empirical pathway of the study are detailed in this section. Specifically, the empirical procedures begin with preliminary tests, and then the main estimation approach.

2.1. Data

The variables used in this study are derived from the existing environmental literature. The purpose of this study is to investigate the validity of the EEC hypothesis in China, which is a major player in the world economy and one of the largest emitters of carbon dioxide. To examine the impact of employment on environmental degradation in China, we use carbon emissions, which represent environmental pollution, as the dependent variable in our model. The independent variables are GDP, employment, renewable energy consumption, fossil energy consumption and material consumption, all of which represent economic growth. We also specifically study the impact of employment on carbon emissions in China's industrial, service and agricultural sectors using these variables. Owing to lack of data on the variables for some years, the dataset from 1990 to 2020 is used in the model analysis. Detailed information and sources of the variables are given in Table 1.

2.2. Model design

Most models used by environmental economists are based on the equation $I = P.A.T$ (Holdren and Ehrlich, 1974; York et al., 2003). Given that this environmental impact model is found suitable for econometric analysis and easily explained (economically) intuitively, I refers to environmental impact/pollution, P is population, A denotes affluence, and T is technology. In this study, the same relationship can be expressed as follows:

$$I = x + zY, \tag{1}$$

where I is pollution, and Y is income/production. Equation (1) shows that there is a positive relationship between environmental impact and production/income. That is, environmental degradation will increase as production/income increases. On the other hand, Okun's law states that unemployment and GDP are closely related (Okun, 1962). It has been empirically established that the relationship between unemployment and GDP is negative. That is,

$$U = a - bY, \tag{2}$$

U is unemployment, and Y is income/production. The above two relationships are modelled by Kashem and Rahman (2020) as follows. Their model (Equation (3)) proposes that environmental pollution and unemployment are negatively related:

$$I = c - dU, \tag{3}$$

U is unemployment, and I is environmental pollution. The three equations above, namely the same-direction relationship between environ-

Table 1
Description and source of the data variables used in the model.

Notation	Variable	Description	Source
lnCO₂	Carbon emissions	CO_2 emissions (kt)	World Bank Data
lnGDP	Gross domestic product	GDP (constant 2017 international $)	World Bank Data
lnEMP	Total employment	Employment to population ratio, 15+, total (%)	World Bank Data
lnRNWC	Renewable energy consumption	Percentage of total final energy consumption	World Bank Data
lnENU	Fossil fuel energy consumption	Percentage of total	World Bank Data
lnIEMP	Employment in the industrial sector	Employment in industry (% of total employment)	World Bank Data
lnSEMP	Employment in the services sector	Employment in services (% of total employment)	World Bank Data
lnAEMP	Employment in the agricultural sector	Employment in agriculture (% of total employment)	World Bank Data

Note. All variables are transformed into the logarithmic form.

mental pollution and production/income in Okun's Law; the inverse relationship between unemployment and production/income; and finally the inverse relationship between unemployment and environmental pollution in Kashem and Rahman's (2020) environment Philips curve (EPC) hypothesis. The negative correlation of environmental pollution strengthens the idea that there may be a positive relationship between employment and environmental pollution. Thus, these three relationships help us hypothesise that environmental degradation and employment are positively related.

$$I = g + hE, \tag{4}$$

E is employment, and I is environmental pollution. In other words, there is a same-direction relationship between environmental pollution and income or production in the environmental literature; unemployment and income/production are negatively related in Okun's Law; and unemployment and environmental pollution are negatively related in Kashem and Rahman's (2020) EPC hypothesis. Finally, it is known that there is a negative relationship between unemployment and employment in economic theory. Given these relations, it is highly likely that a positive relationship exists between employment and environmental pollution. However, a simple regression analysis of employment on environmental pollution, which ignores the traditional variables involved, may lead to a data bias problem and result in incorrect conclusions. Therefore, we consider the traditional EKC hypothesis form to construct our econometric model (Mahmood et al., 2023; Mahmood, 2023; Quito et al., 2023; Mahmood, 2022; Ali et al., 2020; Koyuncu et al., 2021; Danish et al., 2020; Aydın et al., 2019; Beşer et al., 2017; Mercan and Karakaya, 2015; Ulucak and Bilgili, 2018; Acaravcı and Öztürk, 2010; Halicioglu, 2009). The estimable function is modelled as follows:

$$CO_2 = f\left(GDP, EMP, EMP^2, ENU, RNWC\right). \tag{5}$$

This function is converted into the econometric form by taking the logarithm of all variables. By adding the square of employment to this function, the turning threshold of employment (EMP^2) in the model was examined. This is expressed as follows:

$$lnCO2_t = \propto + \beta_0 lnGDP_t + \beta_1 lnEMP_t + \beta_2 lnEMP_t^2 + \beta_3 lnENU_t + \beta_4 lnRNWC_t + \varepsilon_t, \tag{6}$$

$lnCO_{2t}$ is the logarithmic carbon dioxide emissions (environmental pollution), $lnGDP_t$ is the logarithmic GDP, $lnEMP_t$ is the logarithmic employment rate, $lnEMP_t^2$ is the logarithmic employment rate square, $lnENU_t$ is the logarithmic fossil energy consumption, $lnRNWC_t$ is the logarithmic renewable energy consumption, and ε_t is the error term. $\propto$ in Equation (6) is a constant term that expresses technological developments and efficiency. β_0, β_1, β_2, β_3, and β_4 represent the elasticity coefficients of GDP, employment, employment square, energy use and renewable energy consumption, respectively. When Equation (6) is adapted to the threshold regression model approach, the function of our model is shown in Appendix A.

3. Analysis and results

3.1. Result of unit root test

First, we tested whether the variables used in our model were stationary with the help of the augmented Dickey-Fuller (ADF) unit root test, Dickey-Fuller GLS (ERS) unit root test and Lee Strazicich structural break unit root test. Table 2 shows the results of the ADF and ERS unit root tests. The results indicate that the series do not contain a unit root in the first difference. Table 3 shows the structural breaks of the Lee Strazicich unit root test results. All the variables contain at least one structural break, and according to results, the H0 hypothesis that there is a unit root in the break is rejected.

Table 2
Results of unit root tests.

Variables	ADF		Dickey-Fuller GLS (ERS)	
	I(0)	I(1)	I(0)	I(1)
$lnCO_2$	−0.9817	−1.8452**	−0.3762	−2.7076***
lnGDP	−2.1395	−2.4838	−0.9100	−1.7196*
lnENU	−0.8273	−2.8889**	−0.5973	−2.0666**
lnRNWC	−1.7998	−1.4852	−1.1768	−1.8673*
lnEMP	−2.1621	−4.6768***	−0.4052	−4.4856***
lnIEMP	−2.5871	−2.7009*	−0.8686	−1.6751*
lnSEMP	−1.4852	−2.7272**	−1.2533	−2.7959**
lnAEMP	−2.1621	−4.6768***	−0.3971	−4.5256***

Note. ***, ** and * are statistically significant at the 1%, 5% and 10% levels, respectively.

Table 3
Results of Lee Strazicich structural break unit root test.

Variables	Constant	Break date	Result
$lnCO_2$	−4.0496	2004–2014	H0 is rejected
lnGDP	−4.1455	2007–2014	H0 is rejected
lnENU	−3.1993	2000–2009	H0 is rejected
lnRNWC	−4.8874	2005–2014	H0 is rejected
lnEMP	−3.8786	2007–2009	H0 is rejected
lnIEMP	−3.3098	2007–2012	H0 is rejected
lnSEMP	−4.7398	2005–2014	H0 is rejected
lnAEMP	−3.8671	2007–2009	H0 is rejected

Note: Critic values 1%, 5% and 10% levels are −4.0730, −3.5630, −3.2960, respectively.

3.2. Threshold autoregressive regression estimation

After the stationarity test, we examined whether the variables in our model were linear. Table 4 shows that *lnGDP* is not non-linear given the low R-squared value and the statistically significant probability value of the F-statistic. The BDS test does not confirm the non-linearity of the test results. Although these results are weak, the threshold regression model was used because some variables in the model were non-linear.

Threshold regression models consider the asymmetric fluctuations that occur in the conjuncture where the economy is experiencing contraction and expansion periods. For this reason, the threshold regression estimation method was used in the study. The results of the investigation are detailed in this part. Starting from the preliminary tests that include the stationarity and cointegration tests, the results of the threshold autoregressive regression estimation for the outlined models are discussed in this section.

3.2.1. Threshold regression analysis

In the threshold regression model, it is assumed that the independent variable affects the dependent variable non-linearly. These models consider the asymmetric fluctuations that occur in the conjuncture when an economy experiences periods of recession and expansion. For this

Table 4
Results of Tsay and BDS linearity tests.

Variables	Tsay test		BDS test	
	F-statistic (Prob.)	R^2	BDS Statistic	Prob. value
$lnCO_2$	0.970874	0.011649	0.188927	0.0000
lnGDP	0.029311	0.355679	0.205131	0.0000
lnENU	0.802271	0.047442	0.185026	0.0000
lnRNWC	0.928221	0.022060	0.193708	0.0000
lnEMP	0.468582	0.116501	0.207734	0.0000
lnIEMP	0.453454	0.127640	0.203020	0.0000
lnSEMP	0.453235	0.127742	0.193708	0.0000
lnAEMP	0.446789	0.128764	0.207734	0.0000

reason, more realistic estimations are made with regime threshold models, which presume that there are more linear models in dissimilar regime periods in the economy. In threshold regression models, a threshold variable is determined, and it is used to divide the observations in the model into one or more regime periods. As the value of the threshold variable changes, the relationship between the dependent and independent variables may change non-linearly (Wang, 2015). In this study, the threshold autoregressive (TAR) model developed by Tsay (1989) has been used, which assumes that a non-linear model follows a linear approach in dissimilar periods of the regime. The TAR model equation is as follows:

$$Y_t = \alpha + X_t(q_t < \gamma)\,\beta_1 + X_t\,(q_t \geq \gamma)\,\beta_2 + \varepsilon_t. \tag{7}$$

The variable q_t is the threshold variable; γ is the threshold parameter. β_1 and β_2 are the coefficients of the parameter divided into two regimes, and ε_t is the disturbance. We can also write Equation (7) as

$$Y_t = \alpha^i + \beta_1^i Y_{t-1} + \beta_{pi}^i Y_{t-pi} + \varepsilon_t \tag{8}$$

$\gamma_{i-1} < S_{t-d} < \gamma_i \;\; i = 1,2,3,4,5 \ldots, l.$

It is expressed the number of regime l, the threshold variable S_{t-d}, the threshold value γ in equation (8). d shows the delay length of S_{t-d}. The parameter superscript state refers to the regimes in which the variables are located. The steps of the threshold model are as follows (Cao and Tsay, 1992):

(i) p is a non-persistent AR period, and the probable threshold variables are determined.
(ii) We examine whether each threshold variable is linear.
(iii) Then, the most appropriate threshold variable (Y_{t-d}) is selected from the results of the previous steps.
(iv) Threshold values are detected by means of scatter plots for p and d values.
(v) Finally, the model is estimated using the least squares method.

4. Discussion of main results

Table 5 shows the estimated threshold and the corresponding employment rate, carbon emissions and GDP of each regime-dependent variable. The threshold employment estimates of Model 1, Model 2, Model 3 and Model 4 are 1.836387, 0.8540903, 4.073769 and 0.8479538, respectively. These threshold values divide the model into two different regime periods: economic growth and recession.

4.1. Model 1

The results in Table 6 show that the coefficient of the slope is positive and statistically significant as a function of the threshold value in Model 1. According to these results, in Model 1, when the employment rate is below the threshold ($lnEMP\ (-1) < 1.83$), the positive effect of GDP on carbon emissions is 0.1719, whereas when the employment rate is above the threshold ($lnEMP\ (-1) \geq 1.83$), the positive effect of GDP on carbon emissions is 0.1778. In other words, when the employment rate in the previous period ($lnEMP\ (-1) \geq 1.83$) exceeds the threshold (the 2nd regime period), the impact of GDP on carbon emissions increases by about 3.43%. This suggests that energy consumption in China has a statistically significant and positive effect on carbon emissions of 0.95%, while renewable energy consumption and carbon emissions in the previous period have a negative effect of −0.25% and −0.17%, respectively. These results are supported by the Apergis and Ozturk (2015), Rahman and Alam (2021) and Wang (2012) study results in the literature. The results of Model 1 show that employment has a positive impact on carbon emissions of 37.10%, while employment squared has a negative impact on carbon emissions of −38.15%.

Table 5
Threshold results for each regime-dependent variable.

Dependent Variable: $lnCO_2$		
Threshold Variable	Regime 1	Regime 2
***Model 1:* lnEMP (-1)**	$lnEMP(-1) < 1.836387$	$lnEMP(-1) \geq 1.836387$
***Model 2:* $lnCO_2$(-1)**	$lnCO_2(-1) < 0.8540903$	$lnCO_2(-1) \geq 0.8540903$
***Model 3:* lnGDP (-1)**	$lnGDP(-1) < 4.073769$	$lnGDP(-1) \geq 4.073769$
***Model 4:* $lnCO_2$(-3)**	$lnCO_2(-3) < 0.8479538$	$lnCO_2(-3) \geq 0.8479538$

The results of the threshold regression coefficient estimate for Model 1, Model 2, Model 3 and Model 4 are presented in Table 5.

4.2. Model 2

The threshold in Model 2 indicates that the slope coefficient is positive and statistically significant. That is, in the first regime period in Model 2, when the carbon emissions in the previous period are below the threshold ($lnCO_2(-1) < 0.85$), the employment rate increases carbon emissions by 38.35%. In the second regime period, when the carbon emissions of the previous period are above the threshold ($lnCO_2(-1) \geq 0.85$), the employment rate has a similar positive impact on carbon emissions of 38.34%. Overall, the effect of employment rate on carbon emissions decreases by about 0.02% when the carbon emissions in the previous period ($lnCO_2(-1) \geq 0.85$) exceed the threshold. This result shows that our EEC hypothesis cannot be rejected. The fact that employment squared has a negative effect on carbon emissions also supports this hypothesis. Additionally, according to the other results in Model 2, a 1% increase in energy consumption leads to a 0.92% increase in carbon emissions. Additionally, a 1% increase in renewable energy consumption reduces carbon emissions by 0.21%.

4.3. Model 3

Given the threshold value in Model 3, the slope coefficient is positive and statistically significant. In the first regime period in Model 3, when the GDP level of the previous period is below the threshold value ($lnGDP\ (-1) < 4.07$), the employment rate increases carbon emissions by 39.52%. In the second regime period, when the level of GDP in the previous period is above the threshold ($lnGDP\ (-1) \geq 4.07$), the employment rate has a positive effect on carbon emissions of 39.51%. That is, the effect of the employment rate on carbon emissions decreases by about 0.02% when the level of GDP in the previous period ($lnGDP\ (-1) \geq 4.07$) exceeds the threshold. Additionally, employment squared has a negative effect on carbon emissions, which supports our EEC hypothesis. According to other results in Model 3, a 1% increase in energy consumption causes a 0.92% increase in carbon emissions. Further, a 1% increase in renewable energy consumption reduces carbon emissions by 0.18%.

4.4. Model 4

The slope coefficient is positive and statistically significant given the threshold in Model 4. According to these results, in the first regime period in Model 4, energy consumption increases carbon emissions by 0.55% if the carbon emissions are below the threshold in the previous period ($lnCO_2(-3) < 0.84$). In the second regime period, when carbon emissions exceed the threshold of the previous period ($lnCO_2(-3) \geq 0.84$), energy consumption has a positive effect on carbon emissions of 0.54%. That is, if carbon emissions in the previous period ($lnCO_2(-3) \geq 0.84$) exceed the threshold, the effect of energy consumption on carbon emissions decreases by about 0.01%. According to other results in Model 4, a 1% increase in GDP leads to a 1.13% increase in carbon emissions. In addition, a 1% increase in renewable energy consumption reduces carbon emissions by −0.04%. Moreover, a 1% increase in employment in the industrial sector leads to a statistically significant increase in carbon emissions of about 17.08%. Similarly, employment in the agricultural sector has a positive effect of 23.61%; however, employment in the

Table 6
Test results of the threshold regression models.

Dependent variable $lnCO_2$	Model 1		Model 2		Model 3		Model 4	
	Coef.	*t-ist.*	*Coef.*	*t-ist.*	*Coef.*	*t-ist.*	*Coef.*	*t-ist.*
lnGDP	0.1719 **	2.61	0.0184	0.34	0.0669	1.39	1.1374*	1.89
	lnEMP(-1) < 1.83	2.69						
	0.1778 **							
	lnEMP(-1) ≥ 1.83							
lnEMP	37.1049 **	4.21	38.3568 ***	4.04	39.5227 ***	4.14	–	–
			lnCO₂(-1) < 0.85	4.04	*lnGDP(-1)* < 4.07	4.14		
			38.3490 ***		39.5127 ***			
			lnCO₂(-1) ≥ 0.85		*lnGDP(-1)* ≥ 4.07			
lnEMP2	−9.6682 ***	−4.51	−10.1613***	−4.17	−10.4123 ***	−4.13	–	–
LnCO₂(-1)	−0.1736 ***	−2.27					–	–
lnENU	0.9506 ****	10.24	0.9299***	9.48	0.9293***	9.23	0.5528***	3.86
							lnCO₂(-3) < 0.84	3.80
							0.5457***	
							lnCO₂(-3) ≥ 0.84	
lnRNWC	−0.2502 ***	−4.43	−0.2149***	−4.92	−0.1816***	−3.60	−0.0409**	−2.06
lnIEMP	–	–	–	–	–		17.0809*	1.91
lnSEMP	–	–	–	–	–		−11.1109**	−2.23
lnAEMP	–	–	–	–	–		23.6127	1.22
lnIEMP2	–	–	–	–	–		−8.0897*	−1.88
lnSEMP2	–	–	–	–	–		−36.2465**	−2.83
lnAEMP2	–	–	–	–	–		−25.4096	−1.19
Cons.	−38.1596 ***	−4.51	−38.2676***		−39.8015**	−4.33	−1.3981*	−1.18
R-sq	0.94	0.98			0.97		0.98	
F-ist.prob.	0.0000	0.0000			0.0000		0.0000	

Note. ***, ** and * are statistically significant at the 1%, 5% and 10% levels, respectively.

service sector has a negative effect of −11.11%.

4.4.1. Diagnostics

Table 7 summarises the basic diagnostic test results of the three models included in this study. The probability value of the Jarque-Bera statistic is 0.8084, 0.5880, 0.6100 and 0.8586 for Model 1, Model 2, Model 3 and Model 4, respectively. These values are sufficiently large to reject the null hypothesis. These results imply that the residuals within the three models are normally distributed. Harvey's F-statistic and Breusch-Pagan-Godfrey's F-statistic have probability values of 0.2593 and 0.7495 respectively, which suggests that the residuals are not serially related. It also implies that the variance of the errors is constant in all three models and that the model does not suffer from different variances. The results of the Breusch-Godfrey serial correlation LM test show no autocorrelation in all three models. Additionally, the analysis shows that the probability value of the F-statistic of the Ramsey RESET test is around 0.4839, which is significantly greater than 0.05, confirming the correct specification of the model.

Moreover, Fig. 1 shows the results of the CUSUM and CUSUM squared tests for Model 1, Model 2, Model 3, and Model 4. Since both CUSUM and CUSUM squared estimates are within the 5% confidence interval, the results confirm the structural stability of the parameters. That is, the parameters are reliable and stable in the case of a structural break. In conclusion, our investigation shows that the EEC hypothesis cannot be rejected in China in all three models. These results provide clear evidence of the non-linear relationship between employment and pollution. Moreover, with the summary of the result visually illustrated in Fig. 2, employment-environmental degradation nexus hypothesis is established across the sectors except in the service sector.

Importantly, the robustness result with the Difference-in-Differences (DiD) approach which further a diagnostic justification to the threshold regression outcome while allaying the endogeneity problem is presented in Table 8. Given the revelation from the structural break that show breaks in 2007 and 2009, the DiD model is estimated with Dummy1 and Dummy2 representing treatment and control effects, respectively. Dummy1 is constructed by taking the values 0 and 1 before and after the treatment, respectively. Dummy2 is constructed by assigning a value of 1 to the units not in the treatment group and 0 to the units belonging to the treatment group. Also, the interaction variable represents the Dummy of the DiD in the model.[1] The DiD method result showed that we don't find a significant effect the p-value of the interaction coefficient. As indicated in Table 8, the interaction coefficient is negative, indicating that the treatment effect negatively affects the dependent variable. That is, the fact that the coefficient of interaction is statistically insignificant shows that there is no difference before and after the treatment, so there is no endogeneity problem.

5. Conclusion and policy recommendations

In this paper, we propose a new construct called the environmental employment curve (EEC), which asserts the existence of an inverted U-

Table 7
Results of the diagnostic tests.

	Model 1	Model 2	Model 3	Model 4
CUSUM test	Stable	Stable	Stable	Stable
CUSUM sq. test	Stable	Stable	Stable	Stable
Breusch-Godfrey Serial Correlation LM test	0.7501	0.4558	0.4992	0.4624
Heteroskedasticity Test: Breusch-Pagan-Godfrey	0.7494	0.6839	0.7370	0.9790
Heteroskedasticity Test: Harvey	0.2593	0.9985	0.9318	0.7864
Ramsey RESET test	0.4839	0.1365	0.1086	0.8093
Normality Test: Jarque-Bera	0.8084	0.5880	0.6100	0.8586

[1] Difference-in-Differences method result reveals no significant effect of the interaction coefficient. However, the interaction coefficient is negative, indicating that the treatment effect negatively affects the dependent variable. It means that, outside of the period of interest, there is no significant difference between your treatment and control. This is neither necessary nor sufficient, but it is encouraging regarding the fact that your control group is a good control group. This means that there is no statistically significant difference between the means of the two groups can be concluded which suggests that any observed difference could have arisen due to random chance or sampling error. In other words, there is no evidence to suggest that the difference between the two groups is real and not just a fluke.

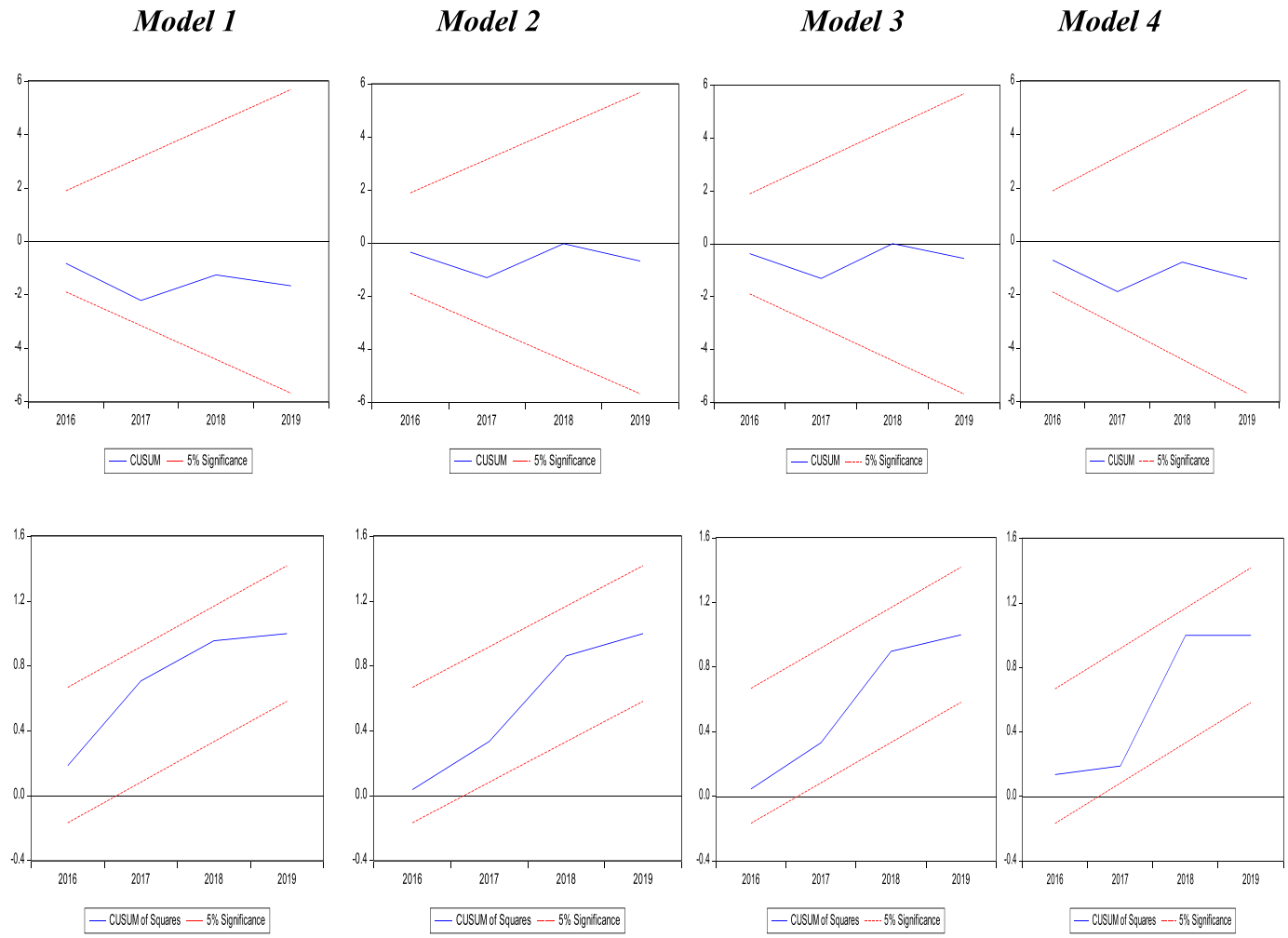

Fig. 1. CUSUM and CUSUMsq stationarity of Model 1, Model 2, Model 3 and Model 4.

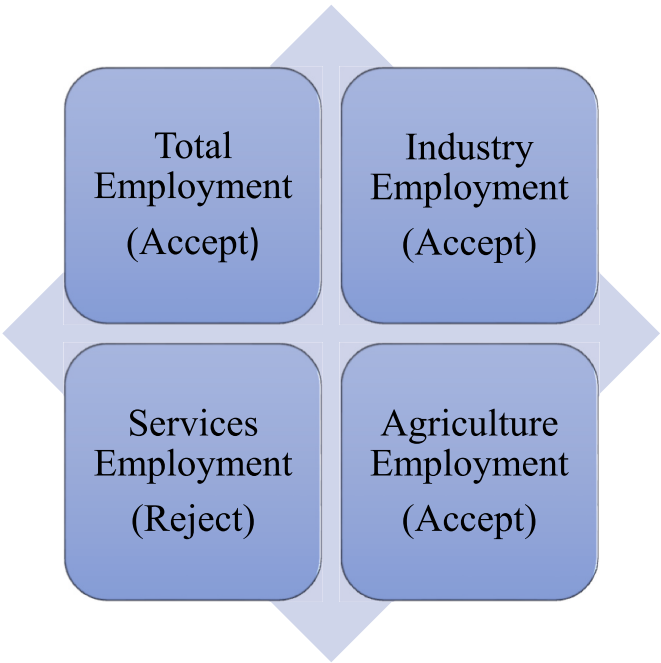

Fig. 2. Summary of ECC hypothesis in China.

Table 8
Robustness result with difference-in-difference estimation.

LCO2	Coefficient	t-statistics	P-value
Dummy 1	0.426	6.680	0.000
Dummy 2	0.038	0.610	0.546
Interaction	-0.077	–0.790	0.437
Intercept	0.408	7.220	0.000
F (3, 26) = 38.96, Prob > F = 0.000 R^2 = 0.818, Adjusted R^2 = 0.707 RMSE = 0.098			

Note: Prob is probability values and RMSE is the Root mean square error.

shaped relationship between environmental pollution and employment rate. According to the EEC, as employment increases up to a certain threshold level, environmental pollution will increase; however, beyond the threshold, environmental pollution will decrease. We tested this concept with help of data from China, which is the world's largest carbon emitter and a country with tremendous employment potential and economic growth. The empirical results supported the EEC hypothesis in all the three studied economic sectors of China. The results showed that with increase in employment, environmental pollution increases up to a certain level and then decreases. We calculated the threshold value at which environmental pollution begins to decrease and found that once the total employment rate reaches 68.13%, carbon emissions begin to decrease. The decrease in emissions beyond a specific threshold also suggests that green technologies and energy transformation can generate employment. Finally, in this study, we also examined the impact of employment on environmental pollution in three Chinese

sectors and found that the EEC hypothesis cannot be rejected in the industrial and agricultural sectors. Specifically, the threshold employment rate for the industrial sector is 53.99% while that for the agricultural sector is 29.89%. Moreover, the results provide evidence that employment in agriculture and industry has a positive effect on carbon emissions, while employment in the service sector has a negative effect. Considering the positive effect of total employment on carbon emissions, it is clearly understood that industrial and agricultural employment has a significant share in this, while employment in the service sector has a negative effect. Moreover, the empirical results show that this turning point will be important for countries to give importance to green energy transformation in their energy policies with increases in welfare levels, as well as the share of the negative impact of services sector employment on carbon emissions. Therefore, it is important for policymakers to take into account the negative effect of service sector employment on carbon emissions. It is expected that the empirical results obtained in this study will contribute to the existing literature by providing a better understanding of the environmental-economic relationship, by presenting a completely new idea developed in the field of environmental economics and by examining the impact of sector-specific employment (industry, services and agriculture) on environmental pollution. Testing the validity of the EEC hypothesis for only China is the limitation of this study. It is believed that testing the validity of EEC hypothesis in different country/groups of countries and different data sets in future investigation will provide a better understanding of the environmental-economic relationship.

5.1. Policy recommendations

In light of the finding that green energy transformation does not result in employment losses, it is important that policymakers remain committed to reducing the use of fossil fuels during the course of a country's energy transition. Apart from green energy transformation, the use of carbon capture and sequestration techniques as well as the transition to electric vehicles can reduce environmental pollution and provide employment opportunities. Further, policies that promote investments in renewable and clean technologies, offer incentives to renewable energy-related institutions and support public-private partnerships in clean energy initiatives should be prioritized. To control environmental pollution, it is important for countries to consider the inverted U-shaped relationship between employment and environmental pollution. Implementing alternative technologies that produce less or no carbon emissions and generate employment without adversely affecting the environment or people's welfare should be a conscious policy choice. Further, since renewable energy consumption reduces carbon emissions, energy efficiency and environment-friendly technologies should be promoted through tax exemptions/reductions, subsidies and investments.

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CRediT authorship contribution statement

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A

Model 1

$$lnCO2_t = \propto + \beta_0 lnEMP_t + \beta_{11} lnGDP_t I(lnEMP_{t-1} < \gamma) + \beta_{12} lnGDP_t I(lnEMP_{t-1} \geq \gamma) + \beta_2 lnEMP_t^2 + \beta_3 lnENU_t + \beta_4 lnRNWC_t + \varepsilon_t.$$

Model 2

$$lnCO2_t = \propto + \beta_0 lnGDP_t + \beta_{11} lnEMP_t I(lnCO2_{t-1} < \gamma) + \beta_{12} lnEMP_t I(lnCO2_{t-1} \geq \gamma) + \beta_2 lnEMP_t^2 + \beta_3 lnENU_t + \beta_4 lnRNWC_t + \varepsilon_t.$$

Model 3

$$lnCO2_t = \propto + \beta_0 lnGDP_t + \beta_{11} lnEMP_t I(lnGDP_{t-1} < \gamma) + \beta_{12} lnEMP_t I(lnGDP_{t-1} \geq \gamma) + \beta_2 lnEMP_t^2 + \beta_3 lnENU_t + \beta_4 lnRNWC_t + \varepsilon_t.$$

Model 4

$$lnCO2_t = \propto + \beta_0 lnGDP_t + \beta_{11} lnENU_t I(lnCO2_{t-3} < \gamma) + \beta_{12} lnENU_t I(lnCO2_{t-3} \geq \gamma) + \beta_2 IEMP_t + \beta_3 lnSEMP_t + \beta_4 lnAEMP_t + \beta_5 lnRNWC_t + \beta_6 lnIEMP_t^2 + \beta_7 lnSEMP_t^2 + \beta_8 lnAEMP_t^2 + \varepsilon_t.$$

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